# Starting Vegetable Seeds Indoors: III. Germination

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## Introduction

This third fact sheet in this series of four guides you through the actual process of starting seeds indoors. The materials you choose may not precisely match those mentioned here. However, the variables and processes discussed are still applicable.

## Step 1: Create the Right Environment

Seeds have some basic requirements that must be met for successful germination. Some of those requirements include the correct temperature, soil moisture, humidity, and light. Create an ideal environment before starting any seeds (Figure 1).

### *Heat*

Most plants you sow will readily germinate at soil temperatures between 65 °F – 75 °F. Some cool-season crops such as broccoli, cabbage, and spinach prefer soil temperatures from 55 °F – 65 °F. For seeds that need higher warmth (e.g., tomatoes, peppers), a relatively inexpensive heating mat can help achieve optimum temperatures if germinating seeds inside a cool room or where temperatures drop too low at night. We suggest coupling heat mats with a thermostat to regulate temperatures. This prevents containers from getting too hot. Fluorescent or other lights emit some heat and may be sufficient if placed within a few inches of the seeds.

### *Humidity*



**Figure 1.***Seed-Starting Space With Heating Pad and Reflective Material to Enhance Light Distribution*

High humidity around seeds helps ensure sufficient water for the seed to imbibe and begin germination. Keep the seed-starting mix evenly moist but not waterlogged. A clear plastic dome placed on top of flats or cell packs is an easy way to maintain moisture. Domes often come in seed starting kits. Alternatively, covering the seeding tray with a large, clear plastic bag achieves the same result. Do not put covered trays in direct sunlight. This may create a greenhouse effect, causing excessively high temperatures to occur.

### *Light*

Some seeds require light to germinate, including many small-seeded plants like snapdragons, petunias, and basil. Seed packages or online crop-specific instructions should indicate light needs. If light is required for germination, use little to no soil when covering the seeds. When in doubt, look up propagation protocols for the crop you are growing. If light is needed for seed light perception or as a heat source, using inexpensive fluorescent shop lights to cover your trays should be sufficient.

## Step 2: Prepare Soil Mix and Fill Grow Containers

Potting soils are often compressed on purpose or during shipping due to stacked bags on pallets. After opening a bag, break up any clumps in the soil you use to fill your containers (Figure 2). Separate the clumps in a separate bin by rubbing the mix between your hands. Dry soils have fine particulate matter that can irritate some respiratory systems; we recommend wearing a dust mask for this step.

Once loosened, you may need to add water if the mix is too dry. Using a pre-wetted mix in growing containers is essential. It makes irrigation more manageable and reduces the risk of seeds washing from containers before they have emerged. The mix should have a similar

moisture level to that of a wrung-out sponge. It is too wet if you can squeeze water from a handful of soil. Incorporate more mix into the bin to obtain the right consistency.

- ***Flats and Similar Containers:*** Fill flats or similarly flat growing containers approximately halfway with the mix. Then, gently lift and tap the containers on the working surface to settle the mix. Do not compress the mix with your hands. Make sure the soilless mix is even and level.
- ***Pots, Pony Packs, and Tray Inserts:*** Add soilless mix and tap containers on the working surface to settle the mix (Figure 2). Keep adding to and settling the mix until it is approximately ¼ inches from the top of the container.



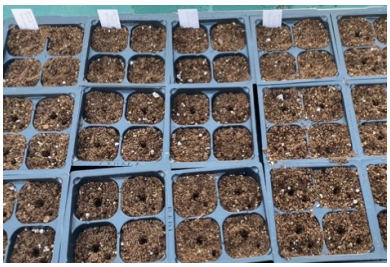


**Figure 2.** *Fill the Cells to the Top With Soilless Mix (left) and Then Tap on a Table to Settle (right)*

## Step 3: Sow the Seeds

Seeds come in different sizes and can have varying requirements for germination. Refer to the seed packet or Utah State University (USU) Extension's fact sheets on the vegetable you are growing. These offer growing instructions. A general recommendation is to plant seeds at a depth of 2–3 times the seed's diameter. There are exceptions to this generalization. Some seeds, such as petunias, require light to germinate. Set these seeds on top of the mix and gently press them in. Before sowing the seeds, label a tag with the crop, variety, and planting date.

- ***Flats and Similar Containers: It is easiest to start the seeds in rows if planting multiple types of seeds in a single open flat. Create shallow furrows by pressing a ruler gently into the mix or dragging a pencil the length of the flat. Space rows 1–2 inches apart. As you place seeds in the soilless mix at the recommended depth and spacing, mark the rows or partial rows with appropriate tags or labels. Cover the seeds with the soilless mix next to the furrow and lightly press onto the seeds to ensure good contact.***
- ***Pots, Pony Packs, and Tray Inserts: Follow the seed packet or fact sheet instructions about how many seeds to place per cell. Make a slight indentation using a pencil or finger in each pot or cell (Figure 3). Press to the depth required by the size of the seed. A tool called a dibble board that presses multiple holes at once can be purchased and used when starting many trays. Plant the seeds and cover them similarly to how you would in open flats. Lightly press on the covered seeds to ensure good seed-to-mix contact.***



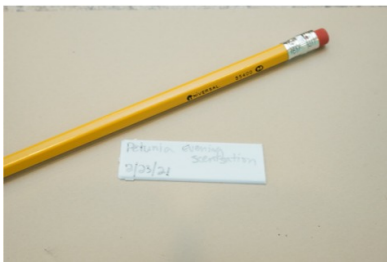


**Figure 3.** *Sow Seeds by Creating a Shallow Indent With a Pencil (left), Planting the Seed, and Labeling Each Unique Seed (right)*

## Step 4: Water

Once the seeds are planted, it is time to saturate the soilless mix with water. One way to do this is by "bottom watering." Fill a basin with a few inches of water and set your seeded container in the water. The seeded container must have drainage holes for this to work. The soilless mix wicks the water from the bottom through the drainage holes. If the mix has not settled sufficiently, water may not reach the top layer of the mix. Use a spray bottle on

a fine mist setting to saturate the top layer (Figure 4). Do not disturb or dislodge the seeds, especially those not covered. When starting seeds on a larger scale or in a greenhouse setting, it is common to use a misting nozzle on a hose to water your newly planted seeds. After the newly planted seeds have been watered, place them in the seed-starting environment you prepared in Step 1. Cover with a humidity dome to retain the moisture in the soilless mix and place them under lights and/or on a heating pad if needed.



**Figure 4.** *Setting the Newly Planted Seeds in a Tray of Water, While Moistening the Top Layer With a Spray Bottle*

*"Bottom watering" allows the soilless mix to wick water from the water-filled tray through drainage holes. Moisten the top layer using a spray bottle set to a fine mist.*

## Step 5: Observe Frequently

Most seed packets give an average range of days for seeds to germinate. Still, seeds can germinate before, during, or after this period. It is essential to check your seeds at least daily to watch for germination and for other issues that may arise (e.g., heating mat/thermostat not working, soilless mix drying out, etc.). Do not keep the mix overly wet, but do not allow dry spots to form before seeds germinate.

## Next Steps

Once your seeds germinate, it's time to care for the seedlings until they are ready to be transplanted into your garden.

## Photos

The authors provided all figure photos.

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